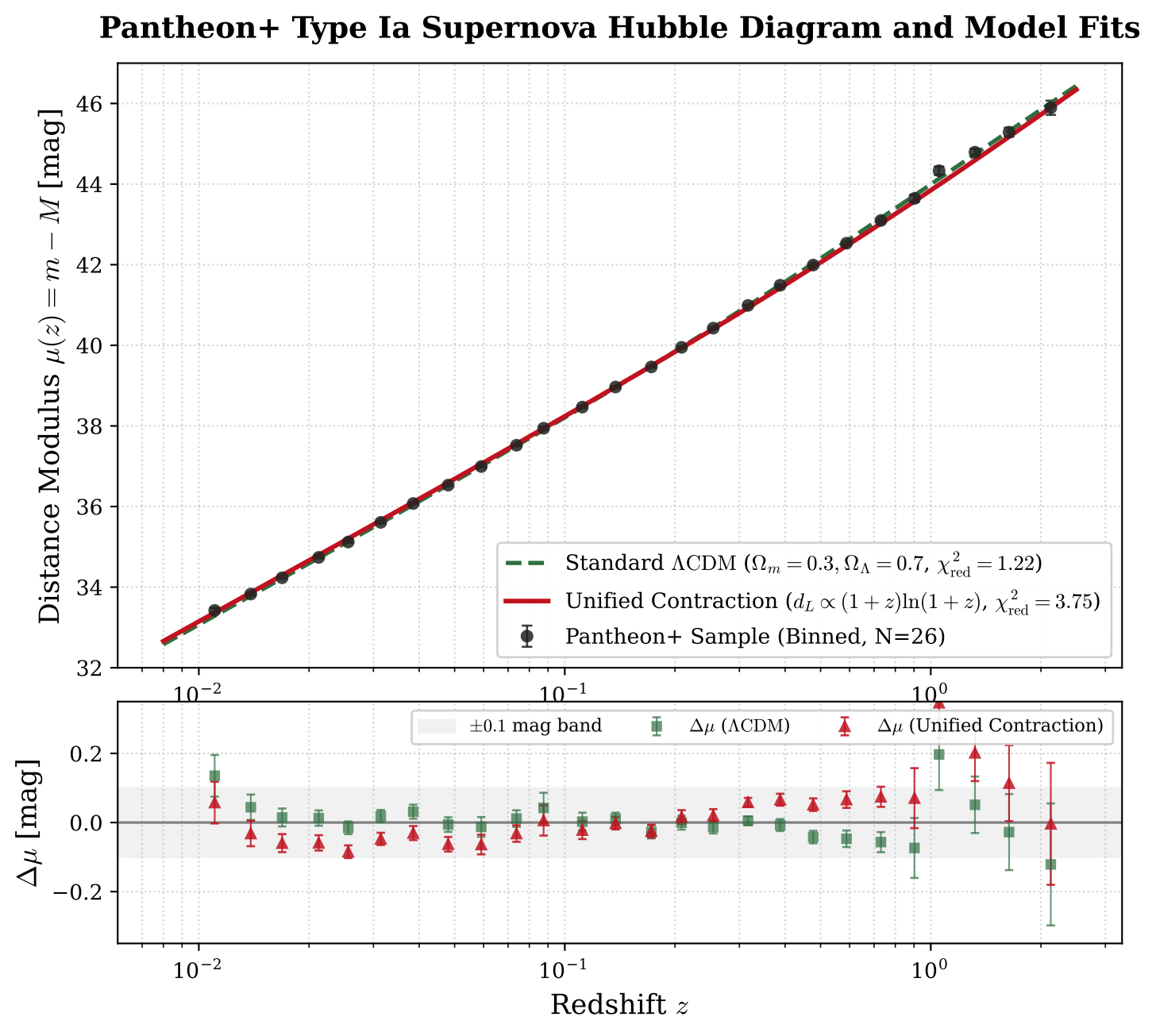

Pantheon+ Type Ia Supernova Hubble Diagram and Model Fits
Distance Modulus $\mu(z) = m - M$ [mag]
Standard ΛCDM ($\Omega_m = 0.3, \Omega_\Lambda = 0.7$, $\chi^2_{\rm red} = 1.22$)
Unified Contraction ($d_L \propto (1+z)\ln(1+z)$, $\chi^2_{\rm red} = 3.75$)
Pantheon+ Sample (Binned, N=26)
± 0.1 mag band
$\Delta\mu$ (ΛCDM)
$\Delta\mu$ (Unified Contraction)
$\Delta\mu$ [mag]
Redshift z